

Metasit Getrak

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Developer with 5+ years of hands-on experience building multi-user mixed-reality applications in Unreal Engine with C++ and OpenXR. Proven track record delivering research-grade XR systems for medical education, Mars exploration simulation, and haptic-feedback surgical training. Skilled in HoloLens 2, Meta Quest 3, hand/eye tracking, and network optimization. Currently completing an M.Sc. in Digital Media (GPA 1.06) at Universität Bremen.

Technical Skills

XR & Engines Unreal Engine (C++), Unity, Godot, OpenXR, HoloLens 2, Meta Quest 3

VR/AR APIs OpenXR, Hand Tracking, Eye Tracking, Phantom Omni Haptics, ROS

Languages C/C++, C#, Python, Java, SQL (MySQL), HTML/CSS, JavaScript

Frameworks NVIDIA Warp, Unreal Network Model, React, Node.js

Tools Git, Visual Studio, VS Code, Linux, Box2D

Networking Unreal multi-user replication, real-time VR network optimisation

Experience

Student Research Assistant — VR/XR Developer (VaMEx3-VTB & DLR School Lab)

CGVR Lab, Universität Bremen · Bremen, Germany

Mar 2024 – Sep 2026

- Optimised VR multi-user network replication for a Mars swarm-exploration simulation (VaMEx3-VTB), reducing latency and improving synchronisation stability across distributed VR clients.
- Built a date/time configuration system in Unreal Engine C++ for mission-critical simulation scheduling.
- Delivered DLR School Lab: a multi-user Mars exploration experience supporting up to 5 concurrent VR users, multiple controllable robots with unique roles, and interactive mission tasks.
- Integrated ROS with Unreal Engine for real-time autonomous swarm data visualisation in VR.

Student Research Assistant — XR Developer (AnatVLab)

Carl von Ossietzky Universität Oldenburg (Remote) · Bremen, Germany

Jan 2025 – Mar 2026

- Ported all HoloLens Anatomy logic from Unreal Blueprints to C++, significantly improving runtime performance and long-term maintainability.
- Extended platform support to Meta Quest 3 using OpenXR, broadening device reach for medical education.
- Optimised system architecture of the HoloLens Anatomy mixed-reality platform.

Student Research Assistant — XR Developer (HoloLens Anatomy)

Carl von Ossietzky Universität Oldenburg (Remote) · Bremen, Germany

Oct 2023 – Dec 2023

- Optimised real-time network communication for the HoloLens Anatomy multi-user MR application.
- Validated cross-continental network connectivity between Thailand and Germany under realistic latency conditions.

Student Research Intern — XR Developer (HoloLens Anatomy)

CGVR Lab, Universität Bremen · Bremen, Germany

Summer 2021, Summer 2022, Summer 2023

- Developed multi-user mixed-reality anatomy application over internet in Unreal Engine targeting HoloLens 2 using OpenXR plugin.
- Implemented the application with region-specific body models and integrated contextual medical information for educational use.
- Collaborated with cross-institutional team at Carl von Ossietzky Universität Oldenburg.

Key Projects

Dental Simulator with Haptic Guidance

Unreal Engine · C++ · OpenXR · Phantom Omni

Mar 2025 – Mar 2026

- Built a VR endodontic surgery trainer with tri-modal guidance (visual, audio, haptic) using Phantom Omni haptic devices for precise surgical tool control.
- Continuation and enhancement of the CGVR Dental Simulator research project.

HoloLens Anatomy / AnatVLab

Unreal Engine · C++ · Unreal Network Model · OpenXR

Jun 2021 – Mar 2026

- Multi-user mixed-reality anatomy platform supporting HoloLens 2 and Meta Quest 3 with free-hand interaction via hand tracking and eye tracking, containing contextual medical information.

DLR School Lab — Mars Game-based Simulation

Unreal Engine · Unreal Network Model · OpenXR

Nov 2024 – Mar 2026

- Mars exploration simulation developed in the Unreal Engine; supports up to 5 VR players and contains multiple tasks for players to solve.

VaMEx3-VTB — Mars Swarm VR

Unreal Engine · C++ · Unreal Network Model · ROS · OpenXR

Mar 2024 – Jun 2026

- VR visualisation and multi-user network layer for an autonomous Mars swarm exploration system; supports both VR and non-VR desktop modes.

DejaView — Differentiable Physics in VR

Unreal Engine · C++ · Python · NVIDIA Warp · OpenXR

Oct 2024 – Mar 2025

- Evaluated two differentiable physics simulators (Uncertain Physics & NVIDIA Warp) with gradient descent, BFGS, and Adam optimisers; visualised results interactively in VR.

Publications

U. Suranin, M. **Getrak**, S. Suebnukarn, P. Haddawy, MS. Yin, R. Weller, P. Aguilar, K. Budsaba, A. Mühlenbrock & G. Zachmann. “Virtual Reality-Based Mirror Guidance for Indirect Vision Training in Minimally Invasive Endodontic Access Cavity Preparation Among Dental Students: Randomized Controlled Crossover Trial.” *JMIR Serious Games* 2026, Vol 14. doi: 10.2196/100884.

M. Prachyabrued, M. **Getrak**, P. Haddawy, P. Sa-Ngasoongsong & G. Zachmann. “Mixed Reality for Group-Based, Integrated Instruction of Anatomy, Pathology, and Physical Examination.” *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW 2026)*, Daegu, Korea, pp. 931–935. doi: 10.1109/VRW70859.2026.00172.

Education

M.Sc. Digital Media (GPA: 1.06 / German scale)

Universität Bremen · Bremen, Germany

Oct 2023 – Jun 2026

- Relevant coursework: Virtual Reality & Physical-Based Simulation · Advanced Computer Graphics · Game Design & Development · Algorithm & Uncertainty: Online Optimization · Media Design

B.Sc. Computer Science (GPA: 3.99 / 4.00)

Mahidol University · Thailand

Aug 2019 – Jul 2023

- Relevant coursework: Computer Graphics · Data Structures & Algorithms · OOP · Web Development · Database · Computer Architecture

Languages

Thai: Native **English:** Upper Intermediate **German:** Basic Communication